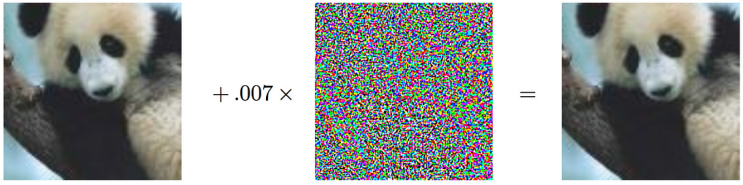
Optimization for Deep Learning (OPT4DL)

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Robustness

Motivation

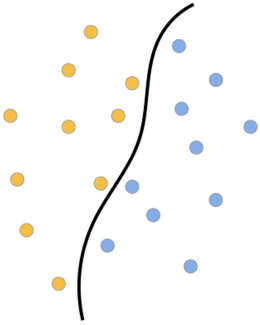
- Our DL models will be used in diverse settings where the conditions of the training process are not necessarily met
- For instance, they may need to withstand changes in the test data compared to training data
  - Change due to bad actors
  - change due to out-of-domain (OOD) Samples
- Thus, we want to induce robustness or resiliency into our training process

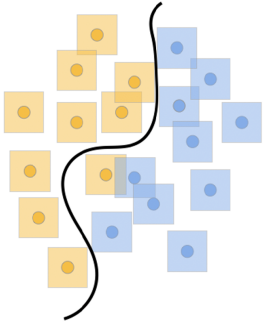


Robustness in Supervised Learning

- Main idea: Reduce sensitivity with respect to changes or perturbations in the data

$$\min_{w} \max_{e:\|e\|\leq\rho} \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i + e) \tag{1}$$





Adversarial Training

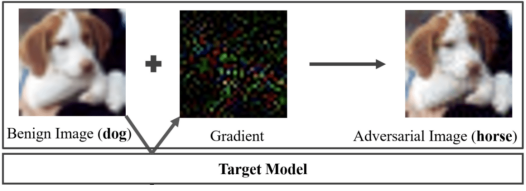
- During the training process, perturb the data according to projected gradient descent:

$$\tilde{z}_t = \text{Proj}\Big(z_t + \tilde{\eta}\nabla_z \ell(w_t, z_t)\Big) \tag{2}$$

- Note: the gradient is w.r.t. the data not parameter w
- the projection aims to ensure to perturbation is not too large and remains bounded
- Then, use the perturbed data $\tilde{z}_t$ (as opposed to the original data z_t) to update the model

$$g_t = \nabla \ell(w_t, \tilde{z}_t), \qquad w_{t+1} = w_t - \eta g_t \tag{3}$$

- Effectively trying to estimate the worst perturbation/attach for each data point



Input Gradient Regularization

- Instead of iteratively solving the max, can we find a closed-form expression for the worst-case perturbation?
- Let us write the first-order Taylor approximation around z

$$\ell(w, z + e) \approx \ell(w, z) + \langle \nabla \ell(w, z), z + e - z \rangle \tag{4}$$

  which will be an affine function of e.
- It turns out the problem below has a closed form solution

$$\arg \max_{e:\|e\| \leq \rho} \ell(w, z) + \langle \nabla_z \ell(w, z), e \rangle = \rho \frac{\nabla_z \ell(w, z)}{\|\nabla_z \ell(w, z)\|} \tag{5}$$

  Leading to the Input gradient regularized robust formulation

$$\min_w \frac{1}{n} \sum_{i=1}^{n} \ell(w, z_i) + \rho \|\nabla_z \ell(w, z_i)\| \tag{6}$$

- Note: Idea similar to sharpness-aware minimization, but there we regularized with gradient w.r.t. w instead

Less Pessimistic Solutions



- Previous approaches based on estimating the worst-cased perturbation could be pessimistic
- A way around is to use random perturbations instead

$$\min_w \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{e\sim p_e}\ell(w, z_i + e)] \tag{7}$$

resembling Gaussian Smoothing (convolution with the PDF of a nice distribution)

- Assume the noise is zero mean and $\mathbb{E}[ee^\top] = 2\rho I_p$. Let us write the second-order Taylor approximation around z

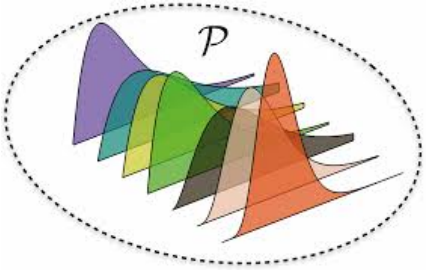
$$\ell(w, z + e) \approx \ell(w, z) + \langle \nabla_z \ell(w, z), e\rangle + \frac{1}{2} e^\top \nabla_z^2 \ell(w, z) e \tag{8}$$

Noting $e^\top \nabla_z^2 \ell(w, z) e = \text{Trace}(e^\top \nabla_z^2 \ell(w, z) e)$ and after using the linearity of trace and expectation leads to

$$\min_w \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i) + \rho \text{Trace}(\nabla_z^2 \ell(w, z_i)) \tag{9}$$

i.e., an input curvature regularization method.

Distributionally Robust Optimization (DRO)



- To induce robustness, the previous approaches were based on perturbation of the input data. Intuitively, the aimed to guess the data distribution during inference. An alternative approach is to directly robustify the changes in the input data distribution:

$$\mathbb{P}(z = z_i) = 1/n \quad \rightarrow \quad \mathbb{P}(z = z_i) = p_i \qquad (10)$$

  where $D(p\|1/n) \leq \rho$.

- Equivalently, if $\mathcal{P}_\rho^D = \{p : D(p\|1/n) \leq \rho\}$ solve the problem

$$\min_w \max_{p \in \mathcal{P}_\rho^D} \mathbb{E}_{z\sim p}[\ell(w, z)]$$

Example: KL-based DRO



- Assume $D(p\|1/n)$ is the KL divergence:

$$D(p\|1/n) = \sum_{i=1}^{n} p_i \log \frac{p_i}{1/n} \tag{11}$$

- After certain steps (crucially using the duality theory), our DRO problem will reduce to

$$\min_w \rho^{-1} \log \Big(\frac{1}{n} \sum_{i=1}^{n} \exp(\rho \ell(w, z_i)) \Big) \tag{12}$$

- Easy to check when $\rho \to 0$, the objective reduces to $\frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i)$ (non-robust) while when $\rho \to \infty$ it reduces to $\max_i \ell(w, z_i)$ (pessimistic robustness)
- Since log is a monotone function, the problem is equivalent to

$$\min_w \frac{1}{n} \sum_{i=1}^{n} \exp(\rho \ell(w, z_i)) \tag{13}$$

  i.e., a new ERM with sample loss $\tilde{\ell}(w, z_i) = \exp(\rho \ell(w, z_i))$.



- Consider the loss prior to removal of log

$$\rho^{-1} \log \left(\frac{1}{n} \sum_{i=1}^{n} \exp(\rho \ell(w, z_i)) \right) \tag{14}$$

- The gradient can be calculated as

$$\frac{\frac{1}{n} \sum_{i=1}^{n} \exp(\rho \ell(w, z_i)) \nabla \ell(w, z_i)}{\frac{1}{n} \sum_{i=1}^{n} \exp(\rho \ell(w, z_i))} = \sum_{i=1}^{n} \alpha_i^{\rho}(w) \nabla \ell(w, z_i), \quad \alpha_i^{\rho}(w) = \frac{\exp(\rho \ell(w, z_i))}{\sum_{j=1}^{n} \exp(\rho \ell(w, z_j))} \tag{15}$$

  i.e., a weighted average of sample gradients where the weights depend on the parameter

- Thus, we can intuitively (but not literally) think of the loss as a weighted ERM where the weights for each sample depend on the parameter w: $\mathbb{P}(z = z_i) = \alpha_i^{\rho}(w)$
- The weights are designed to promote samples with higher loss, thereby inducing robustness